

Addition and Subtraction - Kheyla

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Student

Student: Kheyla

Topic: Addition and Subtraction

Status: completed

Lesson Plan

Learning Objectives

- Calculate the sum and difference of numbers with up to two decimal places using the standard column method with correct decimal point alignment.
- Solve real-world word problems involving money and measurements by applying decimal addition and subtraction operations.
- Model the addition and subtraction of positive and negative integers using a horizontal or vertical number line.
- Compare the results of integer operations within the context of temperature and sea-level changes to verify accuracy.

Lesson Information

Lesson Sections

1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Kheyla! Welcome to our digital classroom. Before we start our global trek today, let's warm up our math muscles with a quick 'Flash Review'. On your screen, you see four boxes. Let's see if we can unlock them together! First, let's look at place value: In the number 4,302, which digit is in the thousands place? Correct! Now, represent the number 2,050 using these virtual 'Base-10' blocks. Excellent. Transitioning to our number games: If I have a sequence 2, 5, 8... what is the next number? Spot on! Finally, solve this quick riddle: I am 10 less than 1,000. Who am I?

Activities:

- *Place Value Identification: Identifying the thousands digit.*
- *Digital Base-10 Block building for 2,050.*

- *Number Pattern Recognition (Arithmetic sequence).*
- *Mental Math subtraction riddle.*

2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Imagine we are traveling from the heat of the Sahara Desert to the frozen peaks of Mount Everest. We need to calculate our budget in dollars (decimals) and track the temperature drops (integers). Math isn't just on paper; it's our survival kit for this trip! Today, we will master how to handle these numbers so we never run out of money or get too cold without knowing it.

Activities:

- *Discussion on travel budgeting and temperature tracking.*

3. Problem Introduction: Decimals (10 minutes)

Teacher Script:

Let's look at our first challenge. We bought a souvenir for \$12.50 and a snack for \$4.75. To find the total, we use 'The Golden Rule of Decimals': Line up the dots! Watch as I stack 12.50 and 4.75. See how the decimal points form a straight line? If we don't line them up, we might accidentally add dollars to cents, and that would be a disaster! Let's try to set up the next one together in the chatbox.

Activities:

- *Demonstration of column addition with decimals.*
- *Visualizing 'phantom zeros' for placeholders (e.g., 12.5 vs 12.50).*

4. Guided Practice: Integers (15 minutes)

Teacher Script:

Now, let's head to the mountains. It's 5 degrees outside, but the sun is going down and the temperature will drop by 8 degrees. On our number line, start at positive 5. Since we are subtracting, we jump to the left. 1, 2, 3, 4, 5... now we hit zero! Keep going... 6, 7, 8. Where are we? Yes, -3! 'Subtracting a positive is like moving left.' Let's try one more: what happens if we are at -2 and the temperature rises by 5?

Activities:

- *Number line jumps for integer addition and subtraction.*
- *Thermometer visualization for 'above/below zero'.*

5. Independent Practice (15 minutes)

Teacher Script:

Kheyla, it's your turn to lead the expedition! I've sent 5 challenges to your screen. Some are about buying gear with decimals, and some are about climbing mountains with integers. I'll be watching your screen, so if you get stuck, just wave! Try to show your work using the column method for the money problems.

Activities:

- Solving 3 decimal word problems (2 addition, 1 subtraction).
- Solving 2 integer problems (elevation shifts).

6. Consolidation & Reflection (5 minutes)

Teacher Script:

You did it! We survived the trip. Today you learned that decimal points are like anchors—they must stay in the same place. You also mastered jumping across zero on the number line. Remember, practicing these keeps your brain sharp for our next adventure! Before you go, don't forget to join <https://learn.algonova.id/login> to see your lesson summary and tasks. What was the most fun part of today's trek?

Activities:

- Summary of 'The Golden Rule of Decimals'.
- Reflection on integer movement.

Homework

Medium Questions:

1. Find the sum: $14.55 + 3.21$
2. Subtract: $25.80 - 12.35$
3. An item costs \$15.75 and you pay with a \$20.00 bill. How much change do you receive?
4. Calculate: $-4 + 9$

Hard Questions:

1. Solve: $124.6 + 8.92$ (Hint: Use a placeholder zero!)
2. The temperature was 3°C and fell by 12°C . What is the new temperature?
3. Calculate the difference: $45.1 - 7.68$
4. A hiker starts at 15 meters below sea level (-15m) and climbs up 40 meters. What is their new elevation?
5. A shop offers a discount of \$4.55 on a \$12.20 shirt. After the discount, you also buy a \$1.99 sticker. What is the final total?
6. Complete the operation: $-15 - (-5)$ (Hint: Subtracting a negative is like adding a positive!)

Answer Key:

- `17.76` ($14.55 + 3.21$)
- `13.45` ($25.80 - 12.35$)
- `4.25` ($20.00 - 15.75$)
- `5` ($-4 + 9$ moves right on number line)
- `133.52` ($124.60 + 8.92$)
- `-9` ($3 - 12 = -9$)

`37.42` ($45.10 - 7.68$)

`25m` ($-15 + 40 = 25$)

`9.64` ($12.20 - 4.55 = 7.65$; $7.65 + 1.99 = 9.64$)

`-10` ($-15 + 5 = -10$)

REVIEW: Nilai Tempat Hingga Ribuan - `6000` (In 6,432, 6 is in the thousands place)

REVIEW: Nilai Tempat Hingga Ribuan - `100` (10 tens make 1 hundred)

REVIEW: Nilai Tempat Hingga Ribuan - `4,052` (4 thousands, 0 hundreds, 5 tens, 2 ones)

REVIEW: Nilai Tempat Hingga Ribuan - `9,999` (The largest 4-digit number)

REVIEW: Diagnostik & Permainan Bilangan - `14` (Next in sequence 2, 5, 8, 11...)

REVIEW: Diagnostik & Permainan Bilangan - `True` (Is $45 > 39$?)

REVIEW: Diagnostik & Permainan Bilangan - `105` ($100 + 5$)

REVIEW: Diagnostik & Permainan Bilangan - `Count by 5s` (What is the pattern in 5, 10, 15, 20?)

Parent Reminder

Today's Math Expedition: Decimals & Positive/Negative Numbers

Kheyla did a great job today! We explored 'The Golden Rule of Decimals' (aligning points for money calculations) and learned how to track temperatures using positive and negative integers. She is becoming quite the global math trekker! Please encourage her to complete the homework on the portal to lock in these new skills.